

Elementary Problems

Introduction

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Introduction

『張三丰當下站起身來，左手持劍，右手捏個劍訣，雙手成環，緩緩抬起，這起手式一展，跟著三環套月、大魁星、燕子抄水、左攔掃、右攔掃……一招招的演將下來，使到五十三式「指南針」，雙手同時畫圓，復成第五十四式「持劍歸原」。張無忌不記招式，只是細看劍招中「神在劍先、綿綿不絕」之意。』

《倚天屠龍記》

In this introduction of the introduction, we give more details of the series of elementary problems.

In short, the series of elementary problems collects problems of different MO topics for beginners to do practices. Its target is beginners since we expect advanced readers should be able to find the suitable problems on their own for practice. Indeed, the IMO shortlist provides over a thousand of problems for that purpose. It is more difficult for beginners to choose some simple problems from many different competitions to work on. Therefore, we hope that this collection can serve as an elementary practice.

This series covers most of the necessary knowledge and skills in various topics in MO. Under each of the four major categories of the IMO, namely, algebra, combinatorics, geometry and number theory, we have classified the problems into different topics. Under each topic, we further group the problems into some sub-topics. This provides more ideas on what the basic things are and what the key ideas are to work on each problem. The problems of each sub-topic are in general ordered by their connections and also the difficulties. Therefore, the ideas for a problem may provide hints to a possible solution to the next problem. However, the problems are only classified in a general sense. A problem can be solved in many different ways and hence one should not limit the thoughts to a single idea.

The problems are elementary in the sense that the skilful MO-ders should find most of them straightforward. This could mean that they may solve the problems at once, may observe the key ideas to solve the problems immediately, or may find that the problems are similar to some well-known problems or results. In particular, there are some standard problems which are known to everyone in the IMO, but could be extremely hard to solve if one has not seen the tricks before. So it is a good chance to learn these tricks through the problems. There are occasionally some difficult problems of IMO level. In most cases, they are included because some of the tricks involved have been used in the previous problems, and so one may also have a try on these problems.

There are suggested solutions provided to the problems. However, they may not be the best solutions. The best solution to each problem is the first solution that one can think of, and hence it should be different for different people. One should always

try to work on the problems instead of just looking at the solutions because this is the only way to improve the ability in MO. So the solutions only serve as a reference.

To provide more insights on various MO topics, we are going to say a few words on the topics one by one in the following. We shall also mention the common and standard tricks involved to solve the problems in each topic.

Algebra

1 Functional Equation

Functional equation is the most common type of algebra problems in the IMO. Most questions ask us to find all functions f satisfying certain equations which are true for all numbers in the domain of f . The only trick is to substitute different values to the given equation, compare different equations obtained and simplify the expressions until we obtain $f(x)$ in terms of x only. One may begin by finding the values of some simple terms like $f(0)$ or $f(1)$. Then one may obtain the relationships between some common terms like $f(x)$, $f(-x)$, $f(2x)$, $f(x^2)$ and $f(f(x))$, etc. Some more useful tricks include replacing x by $-x$ or $f(x)$, swapping x and y , making use of symmetry and using induction to prove some hypothetical results.

Even though the solutions would probably be a nice formula, one should not assume f can be expressed in such a nice way. For example, one can never assume f is a polynomial or it is continuous unless it is stated in the question. Also, one should state clearly which equation is using when doing some substitutions. It is a good habit to label every important equation so that we can refer to them later on.

1.1 Cauchy Equation

The Cauchy equation is the equation $f(x + y) = f(x) + f(y)$. If f is a function defined on the set of rational numbers and it satisfies this Cauchy equation, then there exists a constant c such that $f(x) = cx$ for all x . Note that the same does not apply to real-valued functions. This means there are other solutions which cannot be easily described if f is a real-valued function instead. However, if there are some additional conditions such as f is monotonic, or $f(x) \geq 0$ for any $x \geq 0$, or $f(xy) = f(x)f(y)$ for any x, y , then one can draw the same conclusion (probably with additional information on c).

Sometimes we can prove that a function satisfies the Cauchy equation through various substitutions. Then we can immediately determine the images of the function on the rational numbers. This gives us a convenient way to solve some kinds of functional equation problems.

1.2 Integers and Rationals

Functional equation problems for which the functions are defined on the set of integers or rational numbers can always be solved using induction. One may first guess the solution and try to prove that the formula does hold for all positive integers, and

then all integers, and then also all rational numbers. For rational numbers, we may fix the denominator and use induction on the numerator. This method cannot be generalized to real-valued functions because the set of real numbers is uncountable.

1.3 Injective and Surjective Function

Injective function means a function f for which the images of distinct numbers are distinct. Therefore, if f is injective and $f(x) = f(y)$, then we can conclude that $x = y$. This is also how we can make use of the injectivity. In short, we can cancel the function sign on both sides directly (note that this cannot be done in general). Clearly, this can simplify the expression a lot. The injective condition is probably not given in the question. So one has to prove this using the given information. Usually, one starts by assuming $f(x) = f(y)$ for some x, y , and then compares the equations involving x and y respectively to get some similar properties that x and y should satisfy. From this, one may be able to conclude that $x = y$.

Surjective function means a function f for which the range is the same as the codomain. In many cases, the codomain is the set of real numbers. Then for any real number y , we can find some x such that $f(x) = y$ if f is surjective. If we can show that f is surjective, then we can replace $f(x)$ by any specific value we want. This gives more flexibility when doing substitutions. Again, one has to prove the surjective property by definition. This amounts to rewrite the equation in the form $f(\dots) = t$ where t can take any value in the codomain. In case this can be done, it can usually be done in a simple way.

2 Polynomial

Polynomial is a fundamental topic in algebra. It does not occur that frequently in the IMO, but the basic concepts are needed to solve some other algebra problems, and perhaps also number theory problems. Solving polynomial equations is the most basic stuff. Some other properties of polynomials that one should know include the relation between the roots and coefficients, the behaviour of polynomial growth and the divisibility of polynomials with integer coefficients.

2.1 Identity

It is important to be familiar with the common identities that appear in all mathematics problems. To prove some harder identities, one may try to use induction and the method of partial fractions.

2.2 Root and Vieta's Theorem

Vieta's theorem provides the most useful relationship between the roots and the coefficients of a polynomial. For a quadratic polynomial $ax^2 + bx + c = 0$ where $a \neq 0$, if its roots are α and β , then we have

$$\begin{aligned}\alpha + \beta &= -\frac{b}{a}, \\ \alpha\beta &= \frac{c}{a}.\end{aligned}$$

For a cubic polynomial $ax^3 + bx^2 + cx + d = 0$ where $a \neq 0$, if its roots are α, β, γ , then we have

$$\begin{aligned}\alpha + \beta + \gamma &= -\frac{b}{a}, \\ \alpha\beta + \beta\gamma + \gamma\alpha &= \frac{c}{a}, \\ \alpha\beta\gamma &= -\frac{d}{a}.\end{aligned}$$

These can be generalized to a degree n polynomial. Vieta's theorem can be applied to many problems related to the roots of a polynomial.

Another important result concerning the roots of a polynomial (or any continuous function) is the intermediate value theorem. Let $P(x)$ be a polynomial. If one of $P(a)$ and $P(b)$ is positive and the other is negative, then there exists some c between a and b such that $P(c) = 0$. This result is quite intuitive since we cannot go

from positive to negative without passing through 0. This allows us to confirm the existence of a root without knowing its exact value.

3 Inequality

The popularity of inequality has been greatly declined in the IMO, because many classical inequalities can be proved by various powerful tools. Nevertheless, it is still important to know how to deal with inequalities properly, as it may appear as part of a combinatorial problem (counting, extremal combinatorics, etc.) or a number theory problem. In other words, one has to be familiar with the basic manipulations of inequalities. Always remember to take care of the direction of the inequality sign, and check carefully whether one can do multiplication, squaring, etc.

In terms of the modern type of inequality problems, induction and the simple applications of the AM-GM inequality or Cauchy-Schwarz inequality are among the most common techniques. Therefore, it is still good to go through the classical problems to learn these tricks.

3.1 AM-GM Inequality

The AM-GM inequality states that for any nonnegative real numbers $a_1, a_2, \dots, a_n$, we have

$$\frac{a_1 + a_2 + \dots + a_n}{n} \geq \sqrt[n]{a_1 a_2 \dots a_n}.$$

Equality holds iff $a_1 = a_2 = \dots = a_n$. Although the result is weak, it serves as a simple tool to get a bound for the expression since we can use this inequality to convert between sum and product of some variables. It is in particular useful if we only want to obtain a rough estimation where equality may not hold.

A common nontrivial application of the AM-GM inequality requires one to break down one term into several equal terms intentionally. For example, one may need to rewrite x as $\frac{x}{3} + \frac{x}{3} + \frac{x}{3}$ before applying the inequality. The equality case provides some hints for us to do the partition as each term has to be equal in order that the equality can hold. Working on smaller number of variables may also give us some patterns to guess the solution for the general case.

3.2 Cauchy-Schwarz Inequality

The Cauchy-Schwarz inequality states that for any real numbers $a_1, a_2, \dots, a_n$ and $b_1, b_2, \dots, b_n$, we have

$$(a_1^2 + a_2^2 + \dots + a_n^2)(b_1^2 + b_2^2 + \dots + b_n^2) \geq (a_1 b_1 + a_2 b_2 + \dots + a_n b_n)^2.$$

Equality holds iff $a_i b_j = a_j b_i$ for any $i \neq j$. This inequality has far more applications because there are more flexibilities to choose the variables. It can be used to clear

denominators and radicals in a convenient way. One has to learn these well-known and standard problems in case they may come up as part of an MO problem.

Although there are squares in the expression of the inequality, it does not mean that this can only be used in such kind of inequality problems. This is because we can introduce some square roots in the expressions of a_i and b_j to cancel out this square. We would probably like to do in such a way that each product $a_j b_j$ also has a nice form so that it is easier to proceed.

3.3 Rearrangement Inequality, Chebyshev's Inequality

The rearrangement inequality states that for any real numbers $a_1 \geq a_2 \geq \dots \geq a_n$ and $b_1 \geq b_2 \geq \dots \geq b_n$, and for any permutation $(i_1, i_2, \dots, i_n)$ of $(1, 2, \dots, n)$, we have

$$a_1 b_1 + a_2 b_2 + \dots + a_n b_n \geq a_1 b_{i_1} + a_2 b_{i_2} + \dots + a_n b_{i_n} \geq a_1 b_n + a_2 b_{n-1} + \dots + a_n b_1.$$

Equality holds iff $a_1 = a_2 = \dots = a_n$ or $b_1 = b_2 = \dots = b_n$. Usually, we can order the variables if the inequality is symmetric in the variables. This inequality has basically disappeared in the IMO for many years. Its only application is to prove some very simple and basic inequalities, which can be proved by many other tools as well.

Chebyshev's inequality states that for any real numbers $a_1 \geq a_2 \geq \dots \geq a_n$ and $b_1 \geq b_2 \geq \dots \geq b_n$, we have

$$\begin{aligned} a_1 b_1 + a_2 b_2 + \dots + a_n b_n &\geq \frac{(a_1 + a_2 + \dots + a_n)(b_1 + b_2 + \dots + b_n)}{n} \\ &\geq a_1 b_n + a_2 b_{n-1} + \dots + a_n b_1. \end{aligned}$$

Equality holds iff $a_1 = a_2 = \dots = a_n$ or $b_1 = b_2 = \dots = b_n$. This inequality is powerful in proving some nontrivial inequalities. However, it is in general quite hard to figure out how to rearrange the terms so as to apply the inequality. Also, its style is quite different from modern inequality problems.

3.4 Muirhead's Theorem, Schur's Inequality

Muirhead's theorem states that for any positive real numbers $a_1, a_2, \dots, a_n$ and non-negative real numbers $x_1, x_2, \dots, x_n, y_1, y_2, \dots, y_n$ such that $(x_1, x_2, \dots, x_n)$ majorizes $(y_1, y_2, \dots, y_n)$, we have

$$\sum_{sym} a_1^{x_1} a_2^{x_2} \dots a_n^{x_n} \geq \sum_{sym} a_1^{y_1} a_2^{y_2} \dots a_n^{y_n}.$$

Equality holds iff $a_1 = a_2 = \dots = a_n$ or $(x_1, x_2, \dots, x_n) = (y_1, y_2, \dots, y_n)$. The majorization condition means the sequences satisfy $x_1 \geq x_2 \geq \dots \geq x_n$, $y_1 \geq y_2 \geq \dots \geq y_n$

and

$$\begin{aligned}
x_1 &\geq y_1, \\
x_1 + x_2 &\geq y_1 + y_2, \\
&\dots \\
x_1 + x_2 + \dots + x_{n-1} &\geq y_1 + y_2 + \dots + y_{n-1}, \\
x_1 + x_2 + \dots + x_n &= y_1 + y_2 + \dots + y_n.
\end{aligned}$$

The symmetric sums run over all possible expressions symmetric to the summand. For more details, one is advised to go through some related materials to learn these concepts.

Schur's inequality states that for any nonnegative real numbers a, b, c and $r > 0$, we have

$$a^r(a-b)(a-c) + b^r(b-c)(b-a) + c^r(c-a)(c-b) \geq 0.$$

Equality holds iff $a = b = c$ or one of a, b, c is zero while the other two are equal. We usually use the case $r = 1$, where the inequality can be rewritten as

$$a^3 + b^3 + c^3 + 3abc \geq a^2b + a^2c + b^2a + b^2c + c^2a + c^2b.$$

Schur's inequality is a strong inequality as can be seen from its equality cases.

The two results together can be used to prove most homogeneous symmetric polynomial inequalities in three variables. The solution is straightforward because one only needs to expand everything and applies these results. Thus, it is almost sure that such inequalities will not appear as an IMO problem anymore.

3.5 Jensen's Inequality, Majorization Inequality

Jensen's inequality states that for any real numbers $a_1, a_2, \dots, a_n$ in an open interval (a, b) and $w_1, w_2, \dots, w_n > 0$, and any convex function $f : (a, b) \rightarrow \mathbb{R}$, we have

$$\frac{w_1 f(a_1) + w_2 f(a_2) + \dots + w_n f(a_n)}{w_1 + w_2 + \dots + w_n} \geq f\left(\frac{w_1 a_1 + w_2 a_2 + \dots + w_n a_n}{w_1 + w_2 + \dots + w_n}\right).$$

If f is strictly convex except for a finite number of points, then equality holds iff $a_1 = a_2 = \dots = a_n$. In case f is concave, then the above inequality sign is reversed. The simplest way to check f is convex is to verify $f''(x) \geq 0$ for any $x \in (a, b)$. A simple application of Jensen's inequality is to set the weights w_j 's to be 1. It is useful in proving some inequalities for which the variables are isolated. It is an extremely powerful tool as it has a close relation with calculus.

Majorization inequality states that for any real numbers $x_1, x_2, \dots, x_n, y_1, y_2, \dots, y_n$ with $(x_1, x_2, \dots, x_n) > (y_1, y_2, \dots, y_n)$ and for any convex function f , we have

$$f(x_1) + f(x_2) + \dots + f(x_n) \geq f(y_1) + f(y_2) + \dots + f(y_n).$$

If f is strictly convex except for a finite number of points, then equality holds iff $(x_1, x_2, \dots, x_n) = (y_1, y_2, \dots, y_n)$. In case f is concave, then the above inequality sign is reversed. We usually apply this inequality in combinatorics, where we have some discrete quantities.

3.6 Local Refinement Method

This method means we solve the problem in a few steps in such a way that each time we only focus on one or some of the variables. The other variables are regarded as constants and hence it is easier to find the optimal bound for the expression. The goal of each step is to reduce the number of variables. When doing so, we usually know when the equality should hold and this gives us hint on how each step should be carried out.

3.7 Geometric Inequality (Algebraic)

Sometimes the variables in the inequalities are some quantities related to geometry, in particular triangle geometry. The triangle inequality is one of the tools which may be used. But since the triangle inequality is weak, this is mainly used whenever there is no equality case. For a stronger version, we can use the following substitutions. If a, b, c are the lengths of a triangle, we can define $x = \frac{-a + b + c}{2}$, $y = \frac{a - b + c}{2}$ and $z = \frac{a + b - c}{2}$ so that $a = y + z$, $b = z + x$ and $c = x + y$. This gives an inequality in terms of the positive real variables x, y, z . In this way, we can get rid of the constraints on a, b, c . Note that all other quantities related to the triangle can somehow be expressed in terms of a, b, c . So the inequality obtained is just a usual inequality in three variables.

3.8 Induction

Induction is useful in proving some inequalities involving n variables or inequalities with variables which are positive integers. This tool appears everywhere in mathematics, and so it is important to be familiar with this method. Note that there may be some nontrivial application of induction. For example, we may need to prove a stronger statement so that we can have more information to work on in the inductive step.

4 Sequence

For a sequence problem, sometimes we need to find the general term of the sequence, find the partial sum of the sequence or prove that the sequence satisfies certain properties. One has to know the properties of some basic sequences such as arithmetic sequences, geometric sequences, monotonic sequences, etc. The telescoping series method (meaning to rewrite the expression so that some intermediate terms can be cancelled out when taking the partial sum) and the method of partial fractions are useful when computing the partial sums of some simple sequences.

4.1 Recurrence Sequence

One needs to know how to find the general term of some simple recurrence sequences. In particular, the method of characteristic equations can be used to solve some problems involving a homogeneous linear recurrence sequence with constant coefficients. Such sequences naturally arise from some counting problems in combinatorics. We only state one simple result of this kind. Suppose the sequence $\{a_n\}_{n \geq 1}$ satisfies $a_{n+2} = ta_{n+1} + ua_n$ for all $n \geq 1$ where t and u are some constants with $u \neq 0$. Consider the characteristic equation $\lambda^2 - t\lambda - u = 0$.

- If it has two distinct roots α and β , then there exist constants A and B such that

$$a_n = A\alpha^n + B\beta^n$$

for any $n \geq 1$.

- If it has a double root α , then there exist constants A and B such that

$$a_n = A\alpha^n + Bn\alpha^n$$

for any $n \geq 1$.

The constants A and B can be determined by some initial values such as a_1 and a_2 .

Combinatorics

5 Counting

Counting is required in many combinatorial problems. The basic counting methods include the addition principle, the multiplication principle, the inclusion-exclusion principle and permutation and combination. There is certainly more than one way to do a counting problem. But one has to gain good insights to choose the correct methods to use for each kind of problems. Sometimes a bit of exhaustion cannot be avoided, and so one has to learn how to list out all possibilities in a proper way to count accurately.

5.1 Permutation and Combination

The quantity $P_r^n = n(n-1)\cdots(n-r+1)$ counts the number of ways to choose r objects from n objects in order. The quantity $\binom{n}{r} = \frac{n(n-1)\cdots(n-r+1)}{r!}$ counts the number of ways to choose r objects from n objects not in order. Many combinatorial problems can be rephrased in terms of these two counting problems. Therefore, making use of these quantities helps us a lot when doing counting problems.

5.2 Recurrence Sequence

The method of finding a recurrence relation is useful to solve some counting problems which cannot be solved directly by the addition principle and the multiplication principle. In doing so, we try to rephrase the problem in terms of some discrete quantity n (usually a positive integer) and connect the case n and some smaller cases together. We may express the answer corresponding to the case n in terms of that to the cases $n-1$, $n-2$, etc. Once we have obtained such an algebraic relation (called a recurrence relation), we can find the general term using some algebraic methods to get the answer. This idea is essentially the same as the inductive approach, but we need not propose a formula as the answer in order to proceed.

This method can hardly be replaced by some other simpler methods in most cases. One obvious evidence is that the answer usually involves some powers of radicals, which do not arise naturally in other methods.

5.3 Bijection

The method of bijection means we rephrase a counting problem in terms of a totally different scenario using a nontrivial bijection. The result should be something that is obvious or well-known to us. Sometimes it could be very difficult to find such a bijection even though we know there should be one. For example, if we know that the answer is $n!$, then we can find a mapping from the things we need to count to the permutations of n objects. But it is not guaranteed that there is always a neat mapping. Nevertheless, it is still important to know the well-known bijections to work on some standard problems.

6 Double Counting

Double counting is a tool in combinatorics rather than a type of questions. It means we count the same quantity in two different ways to establish some identities or inequalities (in case we only give an estimation to the quantity). In this way, we are able to prove the existence or non-existence of something, or to establish some useful results. So it is important for us to know more ways to count.

In many cases, we have to count a pair of objects (A, B) . For each fixed object A , we count the number of choices for B , and then sum over all such objects A . Similarly, for each fixed object B , we count the number of choices for A , and then sum over all such objects B . This is the most natural way which allows us to get two different expressions for the same quantity.

6.1 Binomial Coefficient

There are two types of problems in this sub-topic. Firstly, we establish some identities involving binomial coefficients using combinatorial arguments. This means we provide a combinatorial meaning to each side of the identity and give a connection or bijection to show why they are equal. This kind of problems is not popular, but they act as some good exercises in double counting.

Secondly, when we use the method of double counting, we always get an expression involving binomial coefficients. In that case, it is a usual trick to use the majorization inequality (or Jensen's inequality) to get a sharp bound for the expression. Indeed, since the binomial function $\binom{x}{2}$ is convex, whenever $(x_1, x_2, \dots, x_n)$ majorizes $(y_1, y_2, \dots, y_n)$, then we have

$$\binom{x_1}{2} + \binom{x_2}{2} + \dots + \binom{x_n}{2} \geq \binom{y_1}{2} + \binom{y_2}{2} + \dots + \binom{y_n}{2}.$$

The majorization condition is met when, for example, the variables $y_1, y_2, \dots, y_n$ are as close to each other as possible.

7 Existence Problem

An existence problem asks us to show that something satisfying certain combinatorial properties must exist. It is not necessary for us to determine all such things, not even to find one such thing explicitly. This often appears in the topic of extremal combinatorics. The most useful techniques in solving existence problems are the pigeonhole principle and proof by contradiction.

7.1 Pigeonhole Principle

The pigeonhole principle states that if n objects are distributed to m boxes, then there exist $\left\lceil \frac{n}{m} \right\rceil$ objects belonging to the same box. Although the statement is simple, it could be hard to create the objects and boxes and apply the result in an appropriate way. Usually, we may try to do some pairing or grouping to create the boxes, or to make use of some symmetries to do so. There are a few famous problems which can be solved using the pigeonhole principle, and one has to know all of them.

7.2 Proof by contradiction

If we want to show that something must exist, we first assume it does not exist and try to seek for some contradiction. This is the idea of the method of proof by contradiction. In this way, we have more assumptions to work on the problem, which could be convenient to make up the arguments.

8 Extremal Combinatorics

Extremal combinatorics is a common type of problems in combinatorics. It asks us to determine the maximum or minimum value that a quantity could attain under certain combinatorial constraints. Just like an optimization problem in inequalities, we need to prove that the value cannot exceed our proposed optimal value in any case, as well as to show that such an optimal value can be attained in at least one case. Usually, it is much easier to find such an optimal case as such a case may arise from a natural construction or can be guessed by trying the small cases. This provides us some hints on what we are going to prove.

To establish the desired bound, there are various ways that we can do. For example, if it is a counting problem, then we may find an algebraic expression for the quantity to be counted and use some algebraic methods to obtain such a bound. Another way is to use the method of double counting to establish some inequalities. If it is an existence problem, then we may use the methods discussed in the previous section.

8.1 Pigeonhole Principle

As suggested, the pigeonhole principle is useful to handle some existence problems in combinatorics. Hence, this can be used to solve one direction of some problems in extremal combinatorics (while the remaining direction is usually trivial).

8.2 Induction

Just like many other combinatorics problems, induction is a useful tool. One would not lose anything by following an inductive approach, since it means we have more assumptions to use but we need not really use all of them.

9 Operation, Game and Invariance

Mathematical game is another common type of problems in the IMO. It asks us to determine the strategy of one of the players to guarantee a win. The most tricky part is perhaps to guess who is the potential winner and provide a strategy for that side. This can be guessed by looking at some small cases or by playing the game for a few times. However, one should always bear in mind that the proposed winner needs not be the real winner. One has to be flexible when working on a problem.

To provide a winning strategy, one has to think of all possibilities that the opponent may proceed, and give a corresponding action for each possibility. For example, we may pair up the possible moves of the game in some ways so that we know how to do under each scenario. This means we make use of the symmetry of the game. Another way is to give a few moves which work in all cases.

Operations can be viewed as one-person games. We may be asked to prove that a certain scenario can be achieved after finitely many steps or to show that the operations must come to an end, etc. In the former case, we may try to construct the steps explicitly, or use the same techniques as solving existence problems. In the latter case, we may use induction or proof by contradiction. Different combinatorial tools can be combined together.

Invariance is a useful tool to solve some problems involving operations and games, but is not limited to these. So we put this tool under the same topic but may occasionally work on some other problems using invariance.

9.1 Algorithm

For algorithm, it means we give a proposal on how to operate (for operation problems) and how to play (for games). The ideas vary from problem to problem since it depends on the rules we have. Working on the small cases to look for patterns and hints is always the best way to start. The greedy algorithm (meaning to make the best choice at each step) could provide a good hint for us, but is never a rigorous reason that the final outcome must be the best.

9.2 Tiling and Colouring

Tiling problems are concerned about using some given shapes to tile a chessboard. Colouring is the most usual way to solve this kind of problems. It means we may colour the cells with different colours by following some patterns, and then we can show that each shape must cover a certain number of cells in a particular colour.

This always provides useful information to us. Some common colourings include the chessboard colouring (colouring the cells black and white alternately), grouping the cells into different congruence classes (assigning the colour k to cells (i, j) satisfying $i + j \equiv k \pmod{m}$ where m is the number of colours used), or using the same colour for some entire rows or columns. Note that colouring is indeed an application of invariance.

9.3 Invariance

We always try to look for various invariance when working on a problem. For example, when working on an operation problem, we may see what kind of things remains unchanged after each operation. This can be the total number of objects, the parity of a quantity, the difference between two values, or the property that a quantity is monotonic increasing, etc. Sometimes a problem becomes very easy once the correct invariance is observed. And such a proof can hardly be replaced by some other methods since we probably need to make use of such properties in any case.

10 Graph Theory

Some combinatorial problems can be rephrased as problems in graph theory. There are many well-known results than we can use in this topic, and it provides a convenient language for us to give our arguments. The tools used to solve graph theory problems almost cover everything in combinatorics, like counting, pigeonhole principle, induction, double counting, bijection, etc.

10.1 Ramsey-typed Problems

The Ramsey problem is to determine the smallest number $r = R(m, n)$ such that if we colour all edges of a simple complete graph K_r in red and blue, then there must be a red K_m or a blue K_n . There are generalizations of this problem by looking for some other special shapes instead of complete subgraphs. Since it is a kind of existence problems, the pigeonhole principle is the most useful tool.

10.2 Extremal Case

Sometimes we may start by choosing an object associated to a quantity which is optimal. For example, in graph theory, we may choose a vertex with the largest degree, choose a path or cycle which is the longest, or choose a graph which has the smallest number of edges. This allows us to have more information to start with. Note that the tool of considering the extremal case is useful in solving all combinatorial problems.

11 Combinatorial Geometry

Combinatorial geometry problems mainly mean combinatorial problems with a little geometric ingredient. Most knowledge in plane geometry are not required to solve such problems. And the skills involved are basically the usual ones in solving combinatorial problems.

11.1 Geometric Facts

In this sub-topic, we require a bit geometric facts to work on the problems. These may include the triangle inequality, the relations between the side lengths and interior angles of a triangle, the angle sum of polygons, and the relations between angles and circles, etc. No deep knowledge is needed.

11.2 Convex Hull

Convex hull is an important concept in combinatorial geometry. For any finite set S of points in the plane not lying on the same straight line, there exists a smallest convex polygon whose vertices are points in S such that every point in S lies on or inside it. This convex polygon is called the convex hull of S . For any sideline AB of the convex hull, all points in S lie on the same side of the line AB . This is the key property of convex hull.

11.3 Extremal Case

Similar to graph theory, looking for the extremal case is always a useful technique. In this topic, we may choose a shortest segment, a triangle with largest area or a circle with largest radius, etc.

Geometry

12 Angle Chasing

Angle chasing is the most important tools in geometry. This means we try to find the relationships between different angles in the figure. Pairs of equal angles or supplementary angles have a lot of different geometric meanings. For example, they may imply a group of concyclic points, a pair of similar triangles, a pair of parallel lines, etc. Using angles as a medium to link up these geometric configurations is a basic technique.

12.1 Finding Similar Triangles and Concyclic Points

Many simple geometry problems can be solved by finding similar triangles and concyclic points, and making use of these results to chase the angles. By drawing an accurate figures, it is not hard to guess which pairs of triangles in the figure should be similar, and which points should lie on the same circle. There are also many standard configurations arisen from some well-known problems or facts in triangle geometry or circle geometry which consist of similar triangles and concyclic points. One should be familiar with these in order to spot the necessary facts easily.

Similar triangles are often used together with length chasing as well. By expressing the ratio of corresponding sides of a pair of similar triangles in different ways, we may be able to prove other pairs of similar triangles. Also, ratio of side lengths has certain connection with the intercept theorem (which is basically similar triangles), ratio of areas of triangles and the angle bisector theorem, etc. One should always try to link up different things together to obtain more geometric meanings of the expression. When dealing with similar triangles, one should note that not only the vertices of the triangles are corresponding points under the similarity, but also all other meaningful points like midpoints of sides and triangle centres, etc. This gives us an easy way to spot more similarities and hence more angle relations.

Making use of concyclicity is the most useful part in angle chasing since a group of concyclic points contains a lot of information on the angles. Many of these angle relations are nontrivial and can hardly be proved without using the properties of concyclic points. Therefore, one has to be familiar with all properties of concyclic points. Looking for such things is always the key to solve some simple geometry problems.

13 Circle

Circles always appear in a geometric figure. It contains a lot of useful properties that we can apply. In addition to the angle relations of concyclic points, there are many other facts related to the centre, the chords and the tangents of a circle.

13.1 Power

Given a circle Γ with centre O and radius r , and given a point P , the power of P with respect to Γ is defined to be the quantity $OP^2 - r^2$. Note that the power is positive when P lies outside Γ , equals zero when P lies on Γ , and is negative when P lies inside Γ . The power of a point theorem states that if we draw any line through P which meets Γ at two points A and B , then the power of P with respect to Γ is $PA \times PB$. Here, directed lengths are used so that the sign is consistent to our definition.

One simple application of power is that if we draw two lines through P which meet Γ at A, B and at C, D respectively, then $PA \times PB = PC \times PD$. This can also be seen from the pair of similar triangles $\triangle PAC \sim \triangle PDB$ and so it is not an advanced tool. However, it is more convenient for us to work on powers as it is simply a kind of length chasing. Whenever we get an expression like $XY \times XZ$, we may try to think of the power of X with respect to a circle passing through Y and Z . There are many different choices of the circle and we may choose the one that is easiest to use.

Note that when P lies outside Γ , we may take a tangent from P to Γ so that it meets Γ at two points which coincide. In that case, if A is the touching point, then the power is simply PA^2 . This limiting case often appears because there are many problems involving tangents of circles.

13.2 Radical Axis

Given two circles Γ_1 and Γ_2 with different centres, the locus of points P which have the same power with respect to both circles is a straight line which is perpendicular to the line joining the centres of the circles. The straight line is called the radical axis of the circles. In particular, if Γ_1 and Γ_2 meet at two distinct points A and B , then the radical axis is the line AB . This is a nontrivial result and can be used to prove that two lines are perpendicular or three points are collinear (by showing that all of them have the same powers with respect to two fixed circles). When there are many circles in the figure, it is always useful to look at the radical axes of some pairs of circles.

Given three circles with pairwise distinct centres, their three pairs of radical axes are either parallel or concurrent. In the latter case, the intersection point is called the radical centre of the three circles. This property can be used to show that some lines are concurrent.

For some more advanced problems, we may need to use the limiting case for which some circles are degenerate. This means their radii are 0 (note that the power with respect to such circles is still well-defined).

14 Triangle Geometry

Triangle geometry constitutes a large class of problems in plane geometry. These problems start from a triangle and introduce many points in various ways and ask us to prove certain results. Since there is a limited variation of the constructions of the problems, experience is extremely important in this area. One may try to connect problems with similar configurations or constructions together. Results found in other problems may become useful intermediate steps to work on a problem. Also, there are many well-known facts related to triangle centres that one should bear in mind. In this sense, it is the easiest for one to train up the ability of doing geometry problems than problems in other topics. One only needs to spend sufficient amount of time on practices.

14.1 Centroid

Problems related to centroid are less common. The only important fact about centroid is that it cuts a median in the ratio $2 : 1$.

14.2 Circumcentre

Properties related to circumcentre and circumcircle are basically those properties of circles. A nontrivial way to show that O is the circumcentre of an acute $\triangle ABC$ is to show that O lies on the perpendicular bisector of BC and $\angle BOC = 2\angle BAC$ (given that O lies inside $\triangle ABC$).

14.3 Orthocentre and Nine Point Circle

There are far a lot more properties of the orthocentre, particularly those interactions with the circumcircle. One should go through those properties again and again until one is familiar with all those facts. Here, we list out some commonly used facts.

Let H be the orthocentre of $\triangle ABC$ and let O be its circumcentre. Then we have $\angle BAH = \angle OAC$. The reflection of H in the line BC lies on the circumcircle, so is the reflection of H in the midpoint M of BC . Let D be the second intersection point of the circle with diameter AH and the circumcircle. Then D lies on the line HM .

The nine point circle passes through the midpoints of the sides, and the feet of the altitudes. Since it is simply the image of the circumcircle under the homothety with centre H and ratio $\frac{1}{2}$, many of its properties can be deduced from those of the

circumcircle and also the properties of homothety. Therefore, it is not an advanced knowledge.

14.4 Incentre and Excentres

Incentre is another commonly appeared triangle centres. All properties related to tangents are useful here. Another important fact is the following. Let I be the incentre of $\triangle ABC$. The line AI meets the circumcircle again at a point D . Then D is the midpoint of the arc $\widehat{BC}$, and it satisfies $DB = DI$.

Excentres seem to be much more difficult. However, excentres are indeed very much similar to the incentre in many sense. They share a lot of similar properties. Therefore, one may try to rephrase a problem with an excentre as a problem with incentre. But one should know that this will never make the problem easier as they are identical. If all the excentres are involved in the same problem, one may try to re-draw the figure starting from the triangle formed by the three excentres. In that case, the vertices of the original triangle becomes the feet of the altitudes, and the original circumcircle becomes the nine point circle. This may make the problem sound easier.

Number Theory

15 Divisibility

Divisibility is the most basic ingredient in number theory. Modern number theory problems in the IMO only requires basic knowledge in divisibility, perhaps together with a little basic stuff in congruences and arithmetic functions. We need to know how to simplify a relation on divisibility and how to find the greatest common divisor of two expressions.

15.1 Prime Number

Prime numbers are the building blocks of the natural numbers since every positive integer can be expressed uniquely as the product of primes. There are some special properties of prime numbers that allow us to handle divisibility more conveniently. For example, if an integer n divides a prime p , then $n = \pm 1, \pm p$. If a prime p divides a product of integers, then p divides at least one of these integers.

15.2 Size

In this sub-topic, we make use of one important property of divisibility. If an integer a divides another integer b where b is nonzero, then we must have $|a| \leq |b|$. This converts a number theory problem to an algebraic problem. Very often, we can use this inequality to give some upper bounds to the size of some integer variables. Then there are only finitely many possibilities and we can check those cases one by one.

15.3 Infinite Descent and Vieta Jumping

The method of infinite descent is used to show that something does not exist. If we assume such a thing exists, then we try to show that there exists another case satisfying the same property, but corresponds to a smaller parameter (say, the sum of all variables). By repeating the same procedure, we obtain more and more cases for which the parameter becomes smaller and smaller. If the parameter is a non-negative integer, then this yields a contradiction since there is no infinite sequence of nonnegative integers which is strictly decreasing. This shows the desired thing cannot exist. This method is basically identical to proof by contradiction together with induction.

Vieta jumping is a special case of the method of infinite descent. It can be applied to some divisibility problems involving quadratic polynomials. We assume a solution exists and we choose the smallest solution (in some sense). By expressing the divisibility in terms of a quadratic equation, we can obtain another solution to the equation since there are two solutions for a quadratic equation. We may try to prove that this solution satisfies all desired properties as well by using the sum of roots and product of roots. In addition, if we can show that this is a smaller solution, then this yields a contradiction and there should be no solution instead. This method was once a difficult trick in number theory, but now becomes well-known to most people in the IMO.

16 Congruence

Congruence is a convenient tool in number theory to work on remainders. One needs to know how to deal with congruences. Note that congruence equations behave very similar to ordinary equations, but we have to do division carefully. A useful property is the following: if $m \equiv n \pmod{d}$, then $P(m) \equiv P(n) \pmod{d}$ for any polynomial $P(x)$. One also needs to know how to solve some simple congruence equations, and to know whether solutions exist without solving them explicitly.

16.1 Chinese Remainder Theorem

Let $n_1, n_2, \dots, n_k$ be pairwise relatively prime positive integers and let $a_1, a_2, \dots, a_k$ be arbitrary integers. The Chinese remainder theorem states that the system of equations

$$\begin{cases} m \equiv a_1 \pmod{n_1}, \\ m \equiv a_2 \pmod{n_2}, \\ \dots \\ m \equiv a_k \pmod{n_k} \end{cases}$$

has a unique solution modulo $n_1 n_2 \cdots n_k$. There is not any important trick to solve an explicit system of equations in this form. The most important fact is the existence of such a solution. This allows us to construct some integers m with certain congruence conditions. To meet the relatively prime condition, we usually restrict the integers $n_1, n_2, \dots, n_k$ to be powers of distinct primes.

16.2 Euler-Fermat Theorem

Let n be a positive integer and a an integer with $(a, n) = 1$. Then

$$a^{\varphi(n)} \equiv 1 \pmod{n}.$$

Here, $\varphi(n)$ is Euler's totient function. When n is a prime number, this gives the Fermat little theorem. These theorems are useful to handle number theory problems involving powers of integers. They used to be some important facts in the IMO, but is now becoming less popular.

17 Diophantine Equation

Diophantine equation refers to equation with integer variables. It is important to know how to solve this kind of problems. Some standard techniques include considering divisibility and congruences, working out some factorizations, or using algebraic ways to argue that some variables cannot be too large.

17.1 Congruence

We may consider the congruences modulo some small numbers like 2, 3, 4, etc. to obtain some informations on the variables. This can be quite useful for equations involving some small powers of integers. Another way is to work on congruences modulo an expression such as a given variable.

17.2 Size

This method means we try to bound the variables so that only a finite number of cases are left. In that case, we can test the remaining cases one by one. This method can be used if we find that there is a dominating term in the equation (like a term with high power). To do so, we simply need to establish some simple inequalities (which could be rather loose). Sometimes we may order the variables to simplify the arguments in case the equation is symmetric in the variables.

17.3 Greatest Common Divisor

After doing factorizations, we may end up with an expression like $a_1 a_2 \cdots a_k = b$. Then we can consider the greatest common divisors of some pairs of the a_j 's and factorize b accordingly. If we only get a few possible factorizations, then in each such case we can break down the equation into a system of equations which is easy to solve.

18 Arithmetic Function and Floor Function

There are a few well-known functions that may appear in number theory problems. One has to be familiar with their properties.

18.1 Divisor Function

Let $n = p_1^{a_1} p_2^{a_2} \cdots p_t^{a_t}$ be the prime factorization of a positive integer $n > 1$. Then the number of positive divisors of n is

$$d(n) = (a_1 + 1)(a_2 + 1) \cdots (a_t + 1).$$

Also, the sum of all positive divisors of n is

$$\sigma(n) = (1 + p_1 + \cdots + p_1^{a_1})(1 + p_2 + \cdots + p_2^{a_2}) \cdots (1 + p_t + \cdots + p_t^{a_t}).$$

These functions are multiplicative. This means if m and n are relatively prime, then $d(mn) = d(m)d(n)$ and $\sigma(mn) = \sigma(m)\sigma(n)$.

18.2 Euler's Totient Function

Let $n = p_1^{a_1} p_2^{a_2} \cdots p_t^{a_t}$ be the prime factorization of a positive integer $n > 1$. Then the number of positive integers from 1 to n relatively prime to n is

$$\varphi(n) = n \left(1 - \frac{1}{p_1}\right) \left(1 - \frac{1}{p_2}\right) \cdots \left(1 - \frac{1}{p_t}\right).$$

Again, $\varphi(n)$ is multiplicative.

18.3 Floor Function

The floor function $\lfloor x \rfloor$ is the greatest integer not exceeding the real number x . So we always have $x - 1 < \lfloor x \rfloor \leq x$. This allows us to get some close bounds to work on the problem.

19 Exponent

This refers to any number theory problems involving powers of integers.

19.1 Order

The concept of order is useful to solve problems involving exponents. Let n be a positive integer and a an integer with $(a, n) = 1$. Then the order of a modulo n is the smallest positive integer d such that $a^d \equiv 1 \pmod{n}$. Such d must exist by the Euler-Fermat theorem since $a^{\varphi(n)} \equiv 1 \pmod{n}$. The most important property about order is that $a^k \equiv a^m \pmod{n}$ iff $k \equiv m \pmod{d}$, where k and m are nonnegative integers. In particular, we have $a^k \equiv 1 \pmod{n}$ iff $d \mid k$. One also needs to know the orders of some small integers modulo other small integers. This allows us to work on some congruences conveniently.

19.2 Lifting the Exponent Lemma

For any prime p and nonzero integer n , let $v_p(n)$ be the greatest nonnegative integer for which $p^{v_p(n)}$ divides n . There are a few variations of the lifting the exponent lemma. Here, we state one of these.

Let p be an odd prime. Let n be a positive integer and let a, b be distinct integers with $a \equiv b \not\equiv 0 \pmod{p}$. Then $v_p(a^n - b^n) = v_p(a - b) + v_p(n)$.

This lemma allows us to compute the largest exponent of p in the prime factorization of an integer of the form $a^n - b^n$ easily. One common case is the case $b = 1$. There are some similar results for integers of the form $a^n + b^n$, and also for the case $p = 2$.

19.3 Valuation

Valuation is the notation $v_p(n)$ we mention above. A useful property is the following. Let $a_1, a_2, \dots, a_n$ be nonzero integers with sum 0 and let p be a prime number. Then the smallest two integers among $v_p(a_1), v_p(a_2), \dots, v_p(a_n)$ are equal. Given an equation with integer variables, we can examine the exponents of p in each term by using this result. This gives us some constraints on the variables.



「滾滾長江東逝水，浪花淘盡英雄。是非成敗轉頭空。

青山依舊在，幾度夕陽紅。」

《臨江仙》